

Beyond Closed-Set Assumptions: Benchmarking and Adapting Few-Shot Action Recognition for Open-Set Scenarios

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Supplementary Material



Figure 1. Qualitative comparison between the baseline SAFSAR and its adaptation to the open-set scenario with FR-Disc on the NTURGBD dataset (not cherry picked). We represent 4 equidistant frame among the 8 used for inference. We assume that a query is classified as *known* if the known score is above 50% and as *unknown* otherwise. A red label means that the prediction is incorrect, while a green label means that the prediction is correct.

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1. Implementation Details

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SAFSAR 5-way 5-shot Since the authors [36] did not release the code, we implemented this method from scratch. We distributed the VideoMAE model over 4 GPUs. We used the pretrained weights of VideoMAE on the Kinetics-400 dataset provided by [38] for all experiments. We used 5 queries for each episodic task T_i . We fixed the weight λ for the $\mathcal{L}_2$ loss of SAFSAR to 0.1. The values for α_{EOS} and

α_{Disc} are fixed to 0.1 and 1000 respectively.

SAFSAR 5-way 1-shot In this case we did parallel training over 4 GPUs, so we consider 20 queries for each episodic task T_i . All parameters are set as in the 5-shot case, except for the learning rates and the number of iterations.

STRM 5-way 5-shot As for SAFSAR 5-way 5-shot, we trained the STRM model by distributing the feature extractor over 4 GPUs. The value of δ_{Disc} is fixed to 10. As in [38], for each episode we consider $4 \times K$ queries (4 queries for each class) and we accumulate gradients for 16 iterations before updating the parameters. For standard datasets –SSv2, HMDB, UCF– we used the same learning rates and number of iterations reported in the original paper [38].

	SSv2	Diving	NTU	HMDB	UCF
Iterations	50K	30K	15K	10K	10K
LR	$4e^{-6}$	$4e^{-6}$	$4e^{-6}$	$4e^{-6}$	$4e^{-6}$

Table 1. Training iterations and learning rates across datasets for SAFSAR 5-way 5-shot

	SSv2	Diving	NTU	HMDB	UCF
Iterations	50K	30K	15K	10K	10K
LR	$4e^{-7}$	$4e^{-7}$	$4e^{-7}$	$4e^{-8}$	$4e^{-8}$

Table 2. Training iterations and learning rates across datasets for SAFSAR 5-way 1-shot

	SSv2	Diving	NTU	HMDB	UCF
Iterations	75K	75K	40K	20K	20K
LR	$1e^{-3}$	$1e^{-3}$	$1e^{-3}$	$1e^{-4}$	$1e^{-4}$

Table 3. Training iterations and learning rates across datasets for STRM 5-way 5-shot

Support Set Classes:

- Pulling two ends of * so that it gets stretched
- Turning the camera downwards while filming *
- Turning the camera left while filming *
- Moving * and * closer to each other
- Putting * similar to other things that are already on the table

Unknown query: turning the camera upwards while filming *



known score (%)

SAFSAR: 87.76%

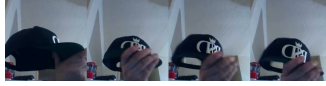
SAFSAR

FR-Disc: 0.72%

Support Set Classes:

- Tipping * with * in it, so * in it falls out
- Moving away from * with your camera
- Putting * similar to each other things that are already on the table
- Pretending to take * out of *
- Putting number of * onto *

Unknown query: holding *



known score (%)

SAFSAR: 87.96%

SAFSAR

FR-Disc: 7.28%

Support Set Classes:

- Plugging * into *
- Moving * and * closer to each other
- Turning the camera upwards while filming *
- Moving * across a surface until it falls down
- Pulling two ends of * so that it separates into two pieces

Unknown query: pretending to take * out of *



known score (%)

SAFSAR: 92.90%

SAFSAR

FR-Disc: 19.47%

Support Set Classes:

- Tilting * with * on it until it falls off
- Pretending to spread air onto *
- Moving * and * closer to each other
- Pouring * onto *
- Spilling * onto *

Known query: pouring * onto *



known score (%)

SAFSAR: 22.42%

SAFSAR

FR-Disc: 88.46%

Figure 2. Qualitative comparison between the baseline SAFSAR and its adaptation to the open-set scenario with FR-Disc on the SSv2 dataset (not cherry picked). We represent 4 equidistant frame among the 8 used for inference. We assume that a query is classified as *known* if the known score is above 50% and as *unknown* otherwise. A red label means that the prediction is incorrect, while a green label means that the prediction is correct.

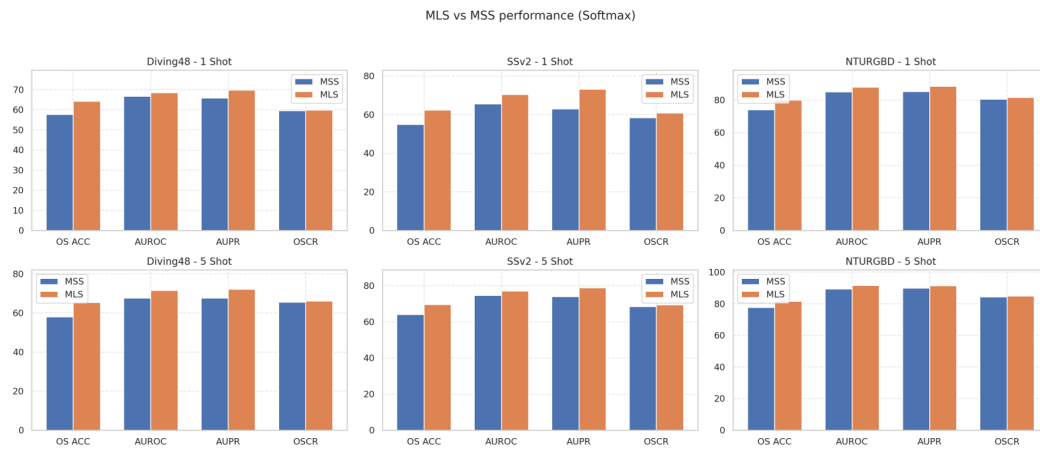


Figure 3. Open-set performances comparison between MLS and MSS for the SAFSAR model.

023 1.1. MLS vs MSS

024 **The importance of not-normalized logits.** In Tab. 4, we
025 report the performance gain of SAFSAR for open-set met-

rics using MLS instead of MSS, where the MLS values are
reported in the paper. We note that the MLS score con-
sistently outperforms the MSS score for implicit methods.

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Dataset	OS-Method	OS ACC		AUROC		AuPR		OSCR	
		1-shot	5-shot	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot
Diving48	Softmax	+6.57	+7.44	+1.80	+3.93	+3.86	+4.54	+0.34	+0.71
	EOS	+3.37	+3.09	+0.04	-0.01	+1.96	+0.96	+0.04	-0.10
SSv2	Softmax	+7.43	+5.57	+4.90	+2.48	+10.05	+4.93	+2.32	+0.90
	EOS	+4.65	+2.28	+0.69	-0.08	+4.62	+2.89	+0.42	+0.30
NTURGBD	Softmax	+5.99	+3.93	+2.83	+2.15	+3.25	+1.40	+0.95	+0.50
	EOS	+3.59	+5.19	+0.58	+2.15	+1.59	+2.40	+0.23	+0.61
HMDB51	Softmax*	+14.44	+22.40	+4.02	+4.10	+3.34	+8.73	+0.44	+1.84
	EOS*	+16.23	+22.48	+5.71	+3.23	+4.67	+8.16	+1.15	+1.57
UCF101	Softmax*	+30.59	+41.25	+1.09	+0.46	+0.79	+1.27	+0.38	+0.34
	EOS*	+31.30	+39.32	+1.29	+0.32	+1.09	+0.70	+0.39	+0.16

Table 4. The gain for open set metrics when using MLS instead of MSS for implicit methods on SAFSAR. Gains are colored: green for positive and red for negative. "*" means that we trained that model for 1K iterations to prevent overfitting.

This confirms the result described in [3, 39] for SAFSAR on spatio-temporal data. Indeed, the SoftMax operation discards the magnitude information that is contained in the logits $\{\hat{s}_{ij}\}_{j=1}^K$.

The value $\hat{u}_i$ can be derived from the logits $\{\hat{s}_i\}_j$ for SAFSAR because this model uses the cosine similarity as similarity function, whose range is in $[-1, 1]$ and can be interpreted directly as probability. We cannot apply the same ablation to STRM because it uses the negative norm of the difference between the query features and the support set classes features, whose range is in $[-\infty, 0]$ and thus can not be interpreted directly as probability.

2. Metrics Definition

2.1. AUROC

AUROC (Area Under the ROC Curve), is defined as:

$$\text{AUROC} = \int_0^1 \text{TPR}(\tau) d\text{FPR}(\tau),$$

where $\text{TPR}(\tau)$ (True Positive Rate) and $\text{FPR}(\tau)$ (False Positive Rate) are computed as:

$$\text{TPR}(\tau) = \frac{\text{TP}(\tau)}{\text{TP}(\tau) + \text{FN}(\tau)}, \quad \text{FPR}(\tau) = \frac{\text{FP}(\tau)}{\text{FP}(\tau) + \text{TN}(\tau)}.$$

and true positives TP are known queries classified as known, false positives FP are unknown queries classified as known, true negatives TN are unknown queries classified as unknown and FN are known queries classified as unknown.

2.2. AuPR

AUPR (Area Under the Precision-Recall Curve) is defined as:

$$\text{AUPR} = \int_0^1 \text{Precision}(\text{Recall}) d\text{Recall},$$

where Precision and Recall are given by:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad \text{Recall} = \text{TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}}.$$

2.3. Closed-Set Accuracy

$$\text{ACC}_{kn}(T_{kn}) = \frac{1}{|T_{kn}|} \sum_{(SS_i, Q_i^{kn}) \in T_{kn}} \mathbb{I}[\hat{y}_i = y_i] \quad (1)$$

where $\hat{y}_i$ is predicted class by $f(SS_i, x_i)$.

2.4. Open-Set Accuracy

We define the open-set accuracy as follows:

$$\text{ACC}^{un}(T) = \frac{1}{|T|} \sum_{(SS_i, Q_i) \in T} \mathbb{I}[\hat{u}_i = u_i] \quad (2)$$

where u_i is the true known/unknown class and $\hat{u}_i$ is the predicted known/unknown class for that sample.

2.5. OSCR

$$\text{CCR}(\tau) = \frac{1}{|T^{kn}|} \sum_{(SS_i, Q_i) \in T^{kn}} \mathbb{I}[\hat{y}_i = y_i] \mathbb{I}[\hat{s}_i > \tau] \quad (3)$$

$$\text{FPR}(\tau) = \frac{\text{FP}(\tau)}{\text{FP}(\tau) + \text{FN}(\tau)} \quad (4)$$

$$\text{OSCR} = \int_0^1 \text{CCR}(\tau) d\text{FPR}(\tau) \quad (5)$$

where τ is a threshold, CCR is the Correct Classification Rate on the P_{FS} problem at threshold τ , and $\text{FPR}(\tau)$ is the false positive ratio at threshold τ . OSCR represents the area under the CCR-FPR curve for each threshold $\tau \in [0, 1]$. This dual evaluation is particularly important for few-shot, where correctly identifying known actions and detecting unknown ones are both crucial requirements. The integral of the metrics that are computed for all threshold –AUROC, AuPR, OSCR – is approximated using the trapezoidal rule.

3. Discriminator Architecture

3.1. SAFSAR

Input shape: 1×768

082	• Linear Layer 768 → 1536
083	• ReLU
084	• Linear Layer 1536 → 768
085	• ReLU
086	• Linear Layer 768 → 64
087	• ReLU
088	• Linear Layer 64 → 1
089	• Sigmoid
090	3.2. STRM
091	Input shape: 28×1152
092	• Linear Layer 1152 → 512
093	• BatchNorm1d
094	• ReLU
095	• Dropout(0.3)
096	• Linear Layer 512 → 256
097	• BatchNorm1d
098	• ReLU
099	• Dropout(0.3)
100	• Linear Layer 256 → 128
101	• BatchNorm1d
102	• ReLU
103	• Dropout(0.3)
104	• Linear Layer 128 → 64
105	• ReLU
106	• Reshape $28 \times 64 \rightarrow 1792$
107	• Linear Layer 1792 → 1
108	• Sigmoid

4. Splits	109
4.1. Diving48	110
Training split	111
Forward_15som_NoTwis_PIKE,	112
Inward_15som_NoTwis_TUCK,	113
Forward_25som_1Twis_PIKE,	114
Back_3som_NoTwis_PIKE,	115
Reverse_35som_NoTwis_TUCK,	116
Forward_25som_NoTwis_TUCK,	117
Forward_45som_NoTwis_TUCK,	118
Reverse_25som_NoTwis_PIKE,	119
Forward_15som_2Twis_FREE,	120
Back_15som_NoTwis_TUCK,	121
Forward_25som_NoTwis_PIKE,	122
Back_35som_NoTwis_TUCK,	123
Reverse_25som_15Twis_PIKE,	124
Reverse_15som_35Twis_FREE,	125
Forward_35som_NoTwis_PIKE,	126
Back_25som_15Twis_PIKE,	127
Inward_35som_NoTwis_TUCK,	128
Inward_15som_NoTwis_PIKE,	129
Back_2som_25Twis_FREE,	130
Back_Dive_NoTwis_PIKE,	131
Back_25som_NoTwis_PIKE,	132
Back_25som_25Twis_PIKE,	133
Inward_25som_NoTwis_PIKE,	134
Reverse_15som_NoTwis_PIKE,	135
Back_15som_15Twis_FREE,	136
Back_25som_NoTwis_TUCK,	137
Forward_1som_NoTwis_PIKE,	138
Back_35som_NoTwis_PIKE,	139
Reverse_15som_05Twis_FREE,	140
Back_3som_NoTwis_TUCK,	141
Back_15som_25Twis_FREE,	142
Back_15som_NoTwis_PIKE	143
Testing split	144
Reverse_15som_15Twis_FREE,	145
Back_Dive_NoTwis_TUCK,	146
Reverse_25som_NoTwis_TUCK,	147
Forward_15som_1Twis_FREE,	148
Reverse_Dive_NoTwis_PIKE,	149
Back_15som_05Twis_FREE,	150
Forward_25som_3Twis_PIKE,	151
Reverse_Dive_NoTwis_TUCK,	152
Back_2som_15Twis_FREE,	153
Inward_25som_NoTwis_TUCK,	154
Reverse_15som_25Twis_FREE,	155
Forward_25som_2Twis_PIKE,	156
Inward_Dive_NoTwis_PIKE,	157
Forward_35som_NoTwis_TUCK,	158
Forward_Dive_NoTwis_PIKE	159

160 **4.2. NTURGBD120**161 **Training split**

162 thumb_down, reach_into_pocket,
163 taking_a_selfie, take_off_glasses,
164 wipe_face, cutting_nails, bounce_ball,
165 phone_call, apply_cream_on_hand,
166 move_heavy_objects, writing,
167 snap_fingers, stand_up, jump_up,
168 thumb_up, side_kick, type_on_a_keyboard,
169 open_a_box, play_magic_cube, reading,
170 juggle_table_tennis_ball, tear_up_paper,
171 put_on_bag, throw, put_on_glasses,
172 arm_swings, shake_head, salute,
173 take_off_bag, kicking_something,
174 throw_up_cap_hat, put_object_into_bag,
175 hush, rub_two_hands, toss_a_coin,
176 apply_cream_on_face, open_bottle,
177 put_on_jacket, cross_toe_touch,
178 ball_up_paper, counting_money,
179 cross_arms, flick_hair, nod_head,
180 make_OK_sign, cross_hands_in_front,
181 shake_fist, take_off_a_shoe, sniff,
182 pick_up, take_object_out_of_bag,
183 drink_water, put_on_headphone,
184 fold_paper, drop, squat_down,
185 take_off_a_hatE

186 **Testing split**

187 hopping, play_with_phone,
188 make_victory_sign, brush_teeth,
189 check_time_from_watch, arm_circles,
190 clapping, point_to_something,
191 put_palms_together, take_off_jacket,
192 take_off_headphone, staple_book,
193 sit_down, run_on_the_spot,
194 shoot_at_basket, put_on_a_hat,
195 brush_hair, butt_kicks, hand_waving,
196 capitulate, put_on_a_shoe, cutting_paper,
197 tennis_bat_swing, eat_meal, cheer_up